# Technical Reference: 1985–1990 RWD Sedan Vehicle Dynamics, UE 5.4/5.5 Async Chaos Physics, and Mono Basin GIS Pipeline

## 1. Vehicle Chassis and Pacejka Tire Model (1985–1990 RWD Sedan)

### Chassis Mechanical Constants

The baseline mechanical parameterization models a late-1980s compact executive rear-wheel-drive (RWD) sedan (E30/W201/R31 platform class) utilizing a front MacPherson strut and rear semi-trailing arm independent suspension. The mass distribution and inertia tensor reflect a longitudinal front-engine, rear-differential layout in which yaw inertia ($I_{zz}$) exceeds pitch inertia ($I_{yy}$) due to overhanging powertrain and fuel tank masses at the vehicle extremities.

| Parameter | Symbol | Value | Unit | Architectural & Dynamic Notes |
|---|---|---|---|---|
| **Total Vehicle Mass** | $m$ | 1,300.0 | $\text{kg}$ | Curb mass plus driver ballast; sprung mass $m_s = 1{,}140.0\text{ kg}$ |
| **Weight Distribution (F / R)** | $W_f$ / $W_r$ | 53.0 / 47.0 | $\%$ | Static axle loads: $F_{z,\text{front}} = 6{,}759\text{ N}$, $F_{z,\text{rear}} = 5{,}994\text{ N}$ |
| **Center of Gravity Height** | $h_{\text{CG}}$ | 0.510 | $\text{m}$ | Measured above ground plane at static ride height |
| **Unsprung Mass per Corner** | $m_{u,f}$ / $m_{u,r}$ | 38.0 / 42.0 | $\text{kg}$ | Mean $40.0\text{ kg/corner}$ ($160.0\text{ kg}$ total; includes hub, brake, wheel, half-shaft) |
| **Roll Moment of Inertia** | $I_{xx}$ | 440.0 | $\text{kg}\cdot\text{m}^2$ | Total vehicle about longitudinal roll axis through CG |
| **Pitch Moment of Inertia** | $I_{yy}$ | 2,150.0 | $\text{kg}\cdot\text{m}^2$ | Total vehicle about lateral pitch axis |

| Parameter | Symbol | Value | Unit | Architectural & Dynamic Notes |
|---|---|---|---|---|
| | | | | through CG |
| **Yaw Moment of Inertia** | I_{zz} | 2,280.0 | \text{kg}\cdot\text{m}^2 | Dynamic index k^2 / (a \cdot b) = 1.03, yielding neutral-to-responsive yaw damping |
| **Wheelbase** | L | 2.620 | \text{m} | CG longitudinal distances: a = 1.231\text{ m} (front), b = 1.389\text{ m} (rear) |
| **Track Width (F / R)** | t_f / t_r | 1.440 / 1.450 | \text{m} | Static stability factor \text{SSF} = \bar{t} / (2 h_{\text{CG}}) = 1.417 |
| **Spring Rate (Front / Rear)** | k_{w,f} / k_{w,r} | 26,000 / 29,000 | \text{N/m} | Effective wheel rates (Coil rates: 32.5 / 48.0\text{ kN/m} at MR_f=0.89, MR_r=0.78) |
| **Damper Rate (Front / Rear)** | c_{w,f} / c_{w,r} | 2,800 / 3,100 | \text{N}\cdot\text{s/m} | Linearized wheel rates (Split Bump/Rebound: F 2,200/3,400; R 2,400/3,800\text{ N}\cdot\text{s/m}) |

The effective wheel rates (26.0\text{ kN/m} front and 29.0\text{ kN/m} rear) establish natural ride frequencies of 1.42\text{ Hz} at the front axle and 1.60\text{ Hz} at the rear axle. This positive front-to-rear frequency gradient prevents pitch oscillation coupling over road undulations by allowing the faster rear suspension cycle to catch up with the front phase lag at highway speeds. Asymmetrical damping ratios (\zeta_{\text{bump}} \approx 0.28, \zeta_{\text{rebound}} \approx 0.44) prevent suspension jacking over high-frequency surface irregularities while controlling transient roll and pitch weight transfer rates during chicane transitions.

## Pacejka MF94 Pure Lateral Slip Coefficients (195/65 R15 at Nominal F_z = 3,200\text{ N})

The pure lateral force formulation follows the Pacejka Magic Formula (MF94) structure F_{y0}(\alpha) = D \sin\left(C \arctan\left(B\alpha - E(B\alpha - \arctan(B\alpha))\right)\right), where \alpha is the slip angle in radians. For implementations evaluating slip angle in degrees, the stiffness factor scales as B_{\text{deg}} = B_{\text{rad}} \times (\pi / 180).

| Surface | Friction ($\mu_y$) | B ($\text{rad}^{-1}$) [B ($\text{deg}^{-1}$)] | C (Shape) | D ($\text{N}$) | E (Curvature) | Cornering Stiffness $C_{F\alpha}$ ($\text{N/rad}$) [$\text{N/deg}$] |
|---|---|---|---|---|---|---|
| **a) Dry Asphalt** | 1.00 | 10.00 [0.1745] | 1.30 | 3,200.0 | -1.00 | 41,600 [726.1] |
| **b) Wet Tarmac** | 0.75 | 10.25 [0.1789] | 1.30 | 2,400.0 | -1.40 | 31,980 [558.2] |
| **c) Loose Gravel** | 0.50 | 8.50 [0.1484] | 1.25 | 1,600.0 | -0.30 | 17,000 [296.7] |
| **d) Hard Snow / Ice** | 0.20 | 9.00 [0.1571] | 1.20 | 640.0 | -2.00 | 6,912 [120.6] |

Across the four surface regimes, the coupling between the stiffness factor $B = C_{F\alpha}/(C \cdot D)$ and curvature factor E dictates limit-handling progression. On wetarmac, hydrodynamic lubrication lowers the sliding friction ceiling ($\mu_y = 0.75$) while cooling the tread compound, slightly elevating B ($10.25\text{ rad}^{-1}$) and driving E more negative (-1.40). Consequently, the tire preserves high linear turn-in stiffness at small slip angles ($\alpha < 2^\circ$) but saturates abruptly with a steeper post-peak drop-off. Conversely, loose gravel exhibits a less negative curvature factor (E = -0.30) and lower stiffness factor ($B = 8.50\text{ rad}^{-1}$) because aggregate shearing builds a deformable soil wedge ahead of the deflected tread blocks, yielding a broad, forgiving slip plateau at elevated slip angles.

## Load Sensitivity and Transient Tire Relaxation Equations

| Physical Relationship | One-Line Mathematical Formulation | Parameter Definitions & Reference Constants |
|---|---|---|
| **1) Load Sensitivity ($D(F_z)$)** | $D(F_z) = \mu_y(F_z) F_z = (a_1 F_{z,\text{kN}} + a_2) F_z = F_z \left[ \mu_0 + p_{Dy2} \left( \frac{F_z - F_{z0}}{F_{z0}} \right) \right]$ | $F_{z0} = 3,200\text{ N}$, $F_{z,\text{kN}} = F_z / 1000$, $a_1 = -0.045\text{ kN}^{-1}$ ($p_{Dy2} = -0.144$), $a_2 = 1.144$ (dry asphalt) |
| **2) Tire Relaxation ($dF_y/dt$)** | $\frac{dF_y}{dt} = \frac{\lvert V_x \rvert}{\sigma_y} \left( F_{y,\text{ss}}(\alpha) - F_y \right), \quad \text{where} \quad \sigma_y = \frac{C_{F\alpha}}{K_y}$ | $\lvert V_x \rvert$: longitudinal ground speed ($\text{m/s}$); $F_{y,\text{ss}}(\alpha)$: steady-state MF94 force ($\text{N}$); $K_y \approx 75,000\text{ N/m}$ ($\sigma_y \approx 0.55\text{ m}$) |

In an RWD sedan, degressive tire load sensitivity ($a_1 < 0$) governs transient yaw balance during longitudinal weight transfer: a 10% vertical load reduction on the rear axle during a mid-corner lift-off maneuver decreases rear peak lateral capacity $D_{\text{rear}}$ by 9%, whereas the corresponding 9% front vertical load increase raises front peak factor $D_{\text{front}}$ by 8%. Simultaneously, integrating the first-order relaxation length differential equation captures the rotating distance required for tread elements to deflect to 1 - 1/e (63.2%) of steady-state lateral force $F_{y,\text{ss}}(\alpha)$. To prevent numerical stiffness instability as forward velocity approaches zero ($\lvert V_x \rvert \to 0$), the integration step clamps $\max(\lvert$

V_x \rvert, V_{\text{min}}) at V_{\text{min}} = 1.5\text{ m/s} or blends into a damped carcass bristle model at low speeds.

# 2. UE 5.4/5.5 Async Physics Tick (120 Hz Chaos)

## Async Force Registration and Application (TSimCallbackObject)

Configuring Unreal Engine 5.4 and 5.5 with bTickPhysicsAsync=True and AsyncFixedTimeStepSize=0.008333 (120\text{ Hz}) executes the Chaos physics solver on a dedicated background thread decoupled from the variable-rate Game Thread. Because direct UObject or UPrimitiveComponent access from the physics thread violates thread safety, custom suspension and tire forces are injected by extending Chaos::TSimCallbackObject and overriding OnPreSimulate_Internal() (or OnPreIntegrate_Internal() when substepping is enabled). Within this callback, the physics-thread particle API (Chaos::FRigidBodyHandle_Internal*) is retrieved via FPhysicsActorHandle->GetPhysicsThreadAPI(), and forces calculated in SI Newtons (\text{kg}\cdot\text{m/s}^2) are multiplied by 100 to match Chaos centimeter units (\text{kg}\cdot\text{cm/s}^2).

```
// Minimal Chaos 120Hz Async Physics Callback (<25 lines)
struct FVehicleAsyncInput : public Chaos::FSimCallbackInput {
    FPhysicsActorHandle ChassisActor = nullptr;
    TArray<TPair<FVector, FVector>, TInlineAllocator<4>>
WheelWorldPosAndForce; // [Pos(cm), Force(kg*cm/s^2)]
    void Reset() { ChassisActor = nullptr;
WheelWorldPosAndForce.Reset(); }
};
class FVehicleAsyncCallback : public
Chaos::TSimCallbackObject<FVehicleAsyncInput,
Chaos::FSimCallbackNoOutput> {
    virtual void OnPreSimulate_Internal() override {
        const FVehicleAsyncInput* In = GetConsumerInput_Internal();
        if (!In || !In->ChassisActor) return;
        if (Chaos::FRigidBodyHandle_Internal* Handle =
In->ChassisActor->GetPhysicsThreadAPI()) {
            const Chaos::FVec3 CoM =
Chaos::FParticleUtilitiesGT::GetCoMWorldPosition(Handle);
            for (const auto& Wheel : In->WheelWorldPosAndForce) {
                Handle->AddForce(Wheel.Value, false);
                Handle->AddTorque(Chaos::FVec3::CrossProduct(Wheel.Key
- CoM, Wheel.Value), false);
            }
        }
    }
};
// Registration (Game Thread): AsyncCallback =
PhysScene->GetSolver()->CreateAndRegisterSimCallbackObject_External<FV
ehicleAsyncCallback>();
```

## Thread-Safe Input/State Double-Buffering Pattern

Marshaling control inputs and simulation telemetry between the Game Thread and the 120\text{ Hz} Chaos Physics Thread combines the lock-free input queue of TSimCallbackObject with an atomic index-swapped double buffer for zero-lock state readback. When the Game Thread frame time exceeds 8.33\text{ ms}, multiple physics ticks execute per rendering frame and GetConsumerInput_Internal() can return nullptr on intermediate steps; consequently, the physics callback persists the most recent input state locally and integrates tire relaxation over GetDeltaTime_Internal().

```
// Lock-Free Game Thread <-> Async Physics Thread Double-Buffer
Pattern
struct FVehicleTelemetryState { FTransform ChassisTM; FVector4f
WheelSlipAngles; uint32 FrameID = 0; };
class FVehicleDoubleBuffer {
    FVehicleTelemetryState Buffers[2];
    std::atomic<uint32> WriteIndex{0};
public:
    void Publish_PT(const FVehicleTelemetryState& NewState) { //
Called at end of OnPreSimulate_Internal()
        const uint32 NextIdx =
(WriteIndex.load(std::memory_order_relaxed) + 1) & 1;
        Buffers[NextIdx] = NewState;
        WriteIndex.store(NextIdx, std::memory_order_release);
    }
    FVehicleTelemetryState Consume_GT() const { // Called in
AActor::Tick() for visual wheel/suspension meshes
        return Buffers[WriteIndex.load(std::memory_order_acquire)];
    }
};
// GT Producer Hook (Bound to FPhysScene::OnPhysScenePreTick):
void UVehicleMovementComp::OnPreTick_GT(FPhysScene* Scene, float
DeltaSeconds) {
    if (FVehicleAsyncInput* OutIn =
AsyncCallback->GetProducerInputData_External()) {
        OutIn->ChassisActor =
UpdatedPrimitive->GetBodyInstance()->GetPhysicsActorHandle();
        OutIn->WheelWorldPosAndForce = ComputeWheelForces_GT(); // Or
pass raw steer/throttle to solve on PT
    }
}
```

| Execution Stage | Thread Context | Engine Hook / API Method | Memory Access & Synchronization Semantics |
|---|---|---|---|
| **Input Production** | Game Thread (GT) | FPhysScene::OnPhysS | Populates lock-free |

| Execution Stage | Thread Context | Engine Hook / API Method | Memory Access & Synchronization Semantics |
|---|---|---|---|
| | | cenePreTick \to GetProducerInputData_External() | input queue; accesses FRigidBodyHandle_External via GetGameThreadAPI() |
| **Physics Consumption** | Physics Thread (PT) | OnPreSimulate_Internal() \to GetConsumerInput_Internal() | Dequeues latest GT input; caches controls if nullptr during multi-step async ticks |
| **Solver Integration** | Physics Thread (PT) | FRigidBodyHandle_Internal::AddForce / AddTorque | Accumulates linear forces and \mathbf{r} \times \mathbf{F} torques around world-space Center of Mass (CoM) |
| **Telemetry Readback** | Game Thread (GT) | FPhysScene::OnPhysScenePos[span_42](start_span)[span_42](end_span)tTick \to Consume_GT() | Reads latest completed physics state via acquire-fenced atomic index or FSimCallbackOutput |

# 3. Mono Basin GIS Data and UE5 Landscape Scaling

## USGS 3DEP 1m LiDAR Metadata & Bounding Box (Mono Lake / Lee Vining / Tioga Pass, CA)

The highway corridor ascending California State Route 120 from the western shoreline of Mono Lake (1,945\text{ m}) through Lee Vining Canyon to Tioga Pass (3,031\text{ m}) and the adjacent Sierra Nevada crest (Mount Dana, 3,980\text{ m}) spans more than 2,000\text{ m} of vertical relief across volcanic, glacio-lacustrine, and granitic terrain. Bare-earth 1-meter Digital Elevation Models (DEMs) are distributed as 32-bit floating-point Cloud-Optimized GeoTIFFs (COG) derived from Quality Level 1 and Level 2 (QL1/QL2) airborne LiDAR surveys.

| GIS Attribute | Specification / Value | Technical Implementation Notes |
|---|---|---|
| **Standardized Dataset Name** | USGS 3DEP 1m Digital Elevation Model (DEM) (API ID: USGS1m) | Distributed via USGS National Map (TNM) & OpenTopography REST API |
| **Source LiDAR Workunits** | USGS_LPC_CA_YosemiteNP_2019_D19 / CA_InyoMono_2020_B20 | Combined NPS/USGS 3DEP & NCALM Mono Basin airborne LiDAR surveys |
| **Native Horizontal CRS** | NAD83(2011) / UTM Zone 11N (EPSG:6340) | Metric planar grid (1.0\text{ m} \times 1.0\text{ m} pixel spacing); legacy UTM EPSG:26911 |
| **Geographic CRS** | NAD83 (EPSG:4269) / WGS84 | Used for REST API |

| GIS Attribute | Specification / Value | Technical Implementation Notes |
|---|---|---|
| | (EPSG:4326) | bounding-box queries (south, north, west, east) |
| **Vertical Datum** | NAVD88 height (Geoid18) (EPSG:5703) | Orthometric elevation in meters (\text{m}); Web Mercator tile caches use EPSG:3857 |
| **Bounding Box (Geographic)** | \text{S: } 37.8500^\circ\text{N}, \text{ N: } 38.1000^\circ\text{N}, \text{ W: } -119.3000^\circ\text{W}, \text{ E: } -118.9000^\circ\text{W} | Encompasses Tioga Pass (37.9110^\circ\text{N}, -119.2579^\circ\text{W}) to Mono Lake (38.0076^\circ\text{N}, -119.0148^\circ\text{W}) |
| **Bounding Box (EPSG:6340)** | X \in [297,650,\; 333,350]\text{ m E}, Y \in [4,191,800,\; 4,218,750]\text{ m N} | Covers 35.7\text{ km} \times 26.95\text{ km}; Lee Vining at 313,680\text{ m E}, 4,203,120\text{ m N} |
| **Key Benchmark Elevations** | Mono Lake: 1,945\text{ m}; Lee Vining: 2,067\text{ m}; Tioga Pass: 3,031\text{ m} | Basin-to-crest elevation span (Z_{\min}=1,935\text{ m} to Z_{\max}=3,985\text{ m}): \Delta Z = 2,050\text{ m} |

## Mathematical Conversion: 32-Bit Float DEM (Meters) to UE5 16-Bit Landscape Z-Scale

Unreal Engine 5 ALandscape encodes terrain elevation as unsigned 16-bit integers (H_{16} \in [0, 65535]) centered at 32768 (Z = 0.0\text{ cm}). Internally, UE5 applies a vertical quantization constant of \frac{1}{128}\text{ cm} (0.0078125\text{ cm}) per integer unit at a default Z-Scale of 100.0, producing an unscaled vertical range of 65535 \times \frac{1}{128}\text{ cm} \approx 512.0\text{ cm} ([-256.0\text{ cm}, +255.9921875\text{ cm}], or 5.12\text{ m}). Mapping a 32-bit floating-point DEM elevation Z_{\text{float}}(x,y) (in meters,) across the regional elevation bounds [Z_{\min}, Z_{\max}] into UE5 world space requires three coupled transformations: \text{Scale}_Z = \frac{(Z_{\max} - Z_{\min}) \times 100.0}{512.0} = (Z_{\max} - Z_{\min}) \times 0.1953125 \quad \left(\text{Exact discrete span: } (Z_{\max} - Z_{\min}) \times \frac{12800.0}{65535.0}\right) H_{16}(x,y) = \text{clamp}\left( \text{round}\left( \frac{Z_{\text{float}}(x,y) - Z_{\min}}{Z_{\max} - Z_{\min}} \times 65535.0 \right), \; 0, \; 65535 \right) \text{Location}_Z = \left( Z_{\min} + \frac{Z_{\max} - Z_{\min}}{2.0} \times \frac{32768.0}{32767.5} \right) \times 100.0 \approx \left( \frac{Z_{\max} + Z_{\min}}{2.0} \right) \times 100.0

| Mono Basin / Tioga Corridor Parameter | Mathematical Symbol | Computed Value | Engineering Impact on Tire/Suspension Raycasts |
|---|---|---|---|
| **Minimum / Maximum Elevation** | Z_{\min} / Z_{\max} | 1,935.0\text{ m} / 3,985.0\text{ m} | Encloses Mono Lake shoreline (1,945\text{ m}) to Mt. Dana summit (3,980\text{ m}) |
| **Total Elevation Span** | \Delta Z = Z_{\max} - Z_{\min} | 2,050.0\text{ m} | 205,000.0\text{ cm} vertical dynamic range |

| Mono Basin / Tioga Corridor Parameter | Mathematical Symbol | Computed Value | Engineering Impact on Tire/Suspension Raycasts |
|---|---|---|---|
| **UE5 Landscape Scale.Z** | \text{Scale}_Z | 400.390625 | Exact: 400.396735 using 12800 / 65535 divisor |
| **UE5 Landscape Location.Z** | \text{Location}_Z | 296,000.0\text{ cm} | Aligns UE5 world Z = 0 with NAVD88 sea level |
| **Vertical Quantization Step** | \delta z = \Delta Z / 65535 | 3.1281\text{ cm} (0.0313\text{ m}) | Induces \pm [span_72](start_span)[span_72](end_span)1.56\text{ cm} stair-stepping on 1\text{ m} horizontal grids |

Quantizing a 2,050\text{ m} vertical span into 16-bit integer space yields a minimum vertical resolution of \delta z = 3.13\text{ cm} per least significant bit across a 100\text{ cm} (1\text{ m}) horizontal vertex grid. Raw heightfield collision queries over this quantization interval introduce discrete surface normal steps of \Delta\theta \approx \arctan(0.0313) = 1.79^\circ, which excite high-frequency unsprung mass chatter at 120\text{ Hz}. To eliminate vertical quantization noise along the paved SR 120 driving surface, the road corridor must be represented by a continuous procedural spline mesh collider or filtered through a 3 \times 3 Gaussian normal-smoothing kernel during the Chaos wheel contact query.

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